1. Introduction to Azure

→ What is Azure ?

- Microsoft Azure is a cloud computing platform that provides a wide range of services to build, deploy, and manage applications and services through Microsoft's global network of data centers.

→ Why Azure ?

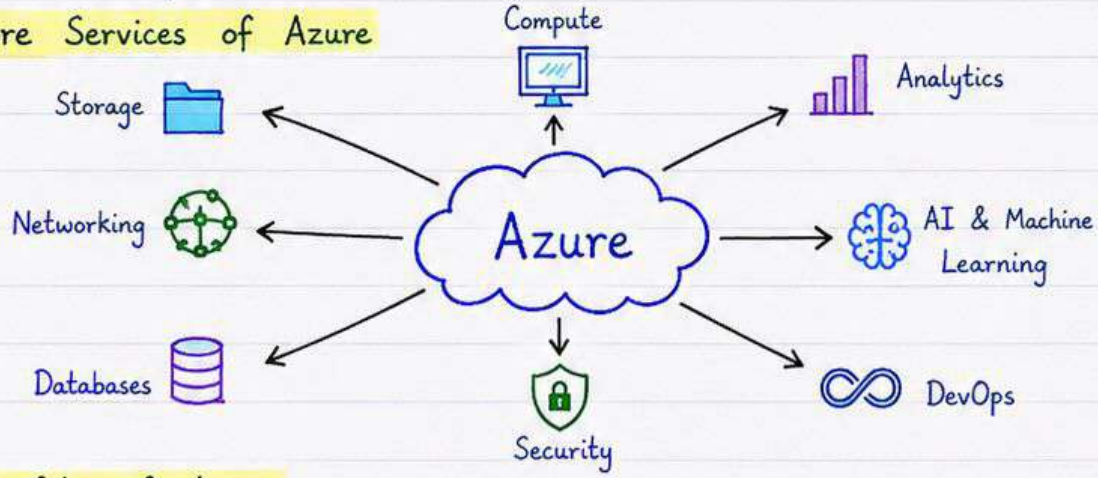
- Highly Available
- Scalable
- Secure
- Cost Effective
- Global Reach
- Hybrid Capabilities
- Large Ecosystem & Community Support

Build, Deploy and Manage anything, anywhere!

→ Key Features of Azure

- Wide range of cloud services
- Global Infrastructure
- High Availability & Scalability
- Security & Compliance
- Cost Management
- Hybrid Cloud Support
- Developer Friendly
- AI & Machine Learning Services
- DevOps & Automation Support

→ Core Services of Azure



→ Benefits of Azure

- ✓ No upfront hardware cost
- ✓ Pay-as-you-go pricing
- ✓ High availability & disaster recovery
- ✓ Best-in-class security
- ✓ Faster time to market
- ✓ Global scale & reach
- ✓ Continuous innovation

2. Azure Core Services (Overview)

→ Azure offers a broad set of cloud services. Below are the major categories and commonly used services.

★ 1. Compute Services

Service	Description	Use Cases
Virtual Machines (VM)	Run Windows or Linux VMs in the cloud.	Lift & shift apps, development, testing.
VM Scale Sets	Automatically scale a set of identical VMs.	High availability apps, load balancing.
Azure Container Instances (ACI)	Run containers without managing servers.	Microservices, quick container deployments.
Azure Kubernetes Service (AKS)	Managed Kubernetes service.	Container orchestration, microservices.
Azure Functions	Serverless compute for event-driven code.	Automate tasks, APIs, background jobs.
App Service	Build, deploy and scale web apps and APIs.	Web apps, REST APIs, mobile backends.

★ 2. Storage Services

Service	Description	Use Cases
Blob Storage	Store large amounts of unstructured data.	Images, videos, backups, logs.
File Storage	Fully managed file shares in the cloud.	Lift & shift apps, shared files.
Queue Storage	Reliable messaging between applications.	Decoupling apps, background tasks.
Table Storage	Store NoSQL key/value data.	Semi-structured data, metadata.
Disk Storage	Persistent block storage for VMs.	OS disks, data disks for VMs.

★ 3. Networking Services

Service	Description	Use Cases
Virtual Network (VNet)	Private network in Azure to connect resources.	Isolated networks, secure communication.
Load Balancer	Distribute incoming traffic across resources.	High availability applications.
Azure DNS	Host DNS domains and manage DNS records.	Domain hosting, name resolution.
VPN Gateway	Secure connection between on-premises and Azure.	Hybrid connectivity, site-to-site VPN.
Azure CDN	Content delivery with low latency globally.	Static content, videos, web apps.

★ 4. Database Services

Service	Description	Use Cases
Azure SQL Database	Managed relational SQL database.	Web SQs, ERP, line-of-business apps.
Azure Cosmos DB	Globally distributed NoSQL database.	Real-time apps, IoT, gaming.
Azure Database for MySQL	Fully managed MySQL database.	WordPress, web apps, CMS.
Azure Database for PostgreSQL	Fully managed PostgreSQL database.	Enterprise apps, analytics apps.
Azure Synapse Analytics	Unified analytics service for big data.	Data warehousing, BI, analytics.

Azure Global Footprint



- 60+ regions worldwide
- Paired regions for high availability
- Millions of customers across different industries
- Data centers on every continent

Service Categories


Compute


Storage


Developer Tools


Security


Networking


Databases


Analytics


Integration

★ 5. Analytics Services

Service	Description	Use Cases
Azure Data Lake Storage	Store and analyze big data in the cloud.	Data lakes, big data analytics.
HDInsight	Managed Hadoop, Spark, Kafka, etc.	Big data processing, ETL workloads.
Stream Analytics	Real-time data streame processing.	IoT analytics, real-time dashboards.
Power BI Service	Business analytics	Reports, dashboards,

★ 6. Security Services

Service	Description	Use Cases
Microsoft Defender for Cloud	Advanced threat protection for cloud workloads.	Security posture, threat detection.
Azure Key Vault	Securely store keys, secrets and certificates.	App secrets, keys, certificate mgmt.
Azure Firewall	Managed cloud firewall service.	Network security, traffic filtering.
Azure Security Center	Unified infrastructure	Security monitoring,

3. Azure Regions, Availability & Pricing

→ Azure Regions

- Azure is available in many regions around the world.
- Each region has one or more data centers.
- Choose a region close to your users for better performance and lower latency.

Example Regions

East US | West Europe | South India | Southeast Asia | Australia East

→ Azure Pricing Models

- Pay-As-You-Go - Pay only for what you use.
- Reserved Instances - Commit for 1 or 3 years and get discounts.
- Spot VMs - Use unused capacity at lower prices.
- Free Tier - Free services and $200 credit for 30 days.
- Price Calculator - Estimate costs before deployment.

→ Azure Service Level Agreement (SLA)

Service Category	Typical SLA
Virtual Machines	99.95%
Storage (LRS)	99.999999999% (11 9's)
SQL Database	99.99%
Azure Cosmos DB	99.99%
App Service	99.95%
Azure Functions	99.95%
Key Vault	99.99%

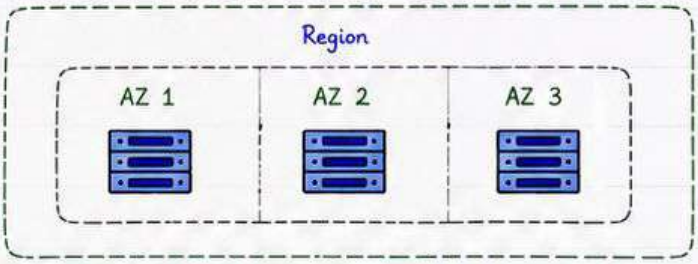
Note: SLA may vary based on region, tier and configuration.

→ Azure Resource Hierarchy

- Management Group → Top level container to manage multiple subscriptions.
- Subscription → Billing boundary. Holds resources and access control.
- Resource Group → Logical container to hold related resources.
- Resources → Individual resources like VMs, Storage, DB, etc.

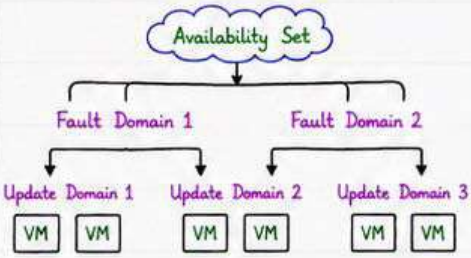
→ Availability Zones (AZ)

- Each region has multiple isolated locations called Availability Zones.
- AZs are physically separated and have independent power, cooling and networking.
- Provide high availability and disaster recovery.



→ Availability Sets (For VMs)

- Ensures VMs are distributed across faults and update domains.
- Helps in managing and updating VMs without downtime.



Fault Domain = Physical hardware rack

Update Domain = Group of VMs updated together

→ Resource Naming Conventions

- Use meaningful, unique and consistent names.
- Helps in managing and identifying resources easily.

Naming Format Example

[Project]-[ResourceType]-[Location]-[Environment]-[Number]

Example:

myproj-vm-eastus-dev-01

→ Best Practices

- ✓ Follow the principle of least privilege.
- ✓ Use Resource Groups for organization.
- ✓ Enable tagging for easy tracking.
- ✓ Monitor and set alerts.
- ✓ Enable backups and disaster recovery.
- ✓ Keep resources updated.



→ Quick Recap



- ✓ Azure provides global cloud services.
- ✓ Choose the right region and services.
- ✓ Ensure high availability and security.

4. Azure Compute Services

→ Azure offers various compute services to run applications in the cloud.

☆ 1. Virtual Machines (VMs)

- IaaS service that provides on-demand, scalable virtual machines.
- Run Windows or Linux workloads in the cloud.
- Full control over OS, storage, networking.

VM Components



VM Usage

- ✓ Lift & shift applications
- ✓ Dev/Test environments
- ✓ Extend on-premises datacenter
- ✓ Disaster recovery
- ✓ Web & App servers
- ✓ High performance computing

☆ 2. Azure App Service

- Fully managed platform to build, deploy and scale web apps, APIs and mobile backends.
- No infrastructure management.
- Supports .NET, Java, Node.js, Python, PHP, etc.

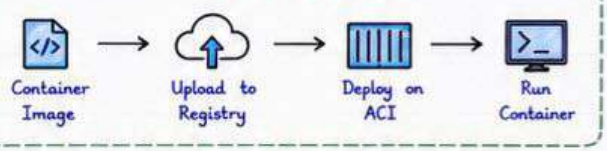
App Service Plan Tiers

Free (F1)	Shared (D1, B1, S1)	Premium (P1v2, P2v2)
For small apps, limited resources.	Cost-effective, shared resources.	High performance, isolated resources.

☆ 3. Azure Container Instances (ACI)

- Run containerized applications without managing servers.
- Lightweight, simple and fast.
- Pay per second - only for what you use.

How ACI Works



☆ 4. Azure Kubernetes Service (AKS)

- Managed Kubernetes service to deploy, manage and scale containerized applications.
- Supports auto-scaling, self-healing and rolling updates.
- Integrates with Azure AD, Monitoring, Networking, etc.

AKS Benefits

- ✓ Fully managed control plane
- ✓ Auto-scaling & self-healing
- ✓ High availability
- ✓ Integrated security
- ✓ Hybrid & multi-cloud support

☆ 5. Azure Functions

- Serverless compute service that runs code in response to events.
- No infrastructure to manage.
- Pay only for the time your code runs.

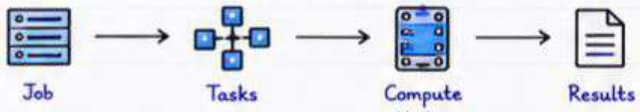
Triggers



☆ 6. Azure Batch

- Execute large-scale parallel and HPC workloads.
- Ideal for batch jobs, simulations, rendering, etc.
- Automatically scales the number of compute nodes.

Batch Workflow



☆ 7. Azure Virtual Desktop (AVD)

- Provide virtual desktop and application delivery in the cloud.
- Secure remote access from anywhere.
- Manage Windows 10/11 multi-session desktops.



Compute Services - Quick Comparison

Service	Type	Best For	Management
Virtual Machines	IaaS	Full control, custom OS & apps	You manage OS & apps
App Service	PaaS	Web apps, APIs, mobile backends	Platform managed
Container Instances	PaaS	Run containers quickly	Platform managed
AKS	PaaS	Kubernetes-based container apps	Platform managed
Azure Functions	Serverless	Event-driven code execution	Fully managed
Azure Batch	PaaS	Large scale batch processing	Platform managed
Azure Virtual Desktop	IaaS/PaaS	Virtual desktops & remote access	Platform managed

Key Takeaway

Choose the right compute service based on your workload type, control needs,

5. Azure Storage Services

→ Azure provides durable, highly available and massively scalable storage solutions for all types of data.

☆ 1. Types of Storage Accounts

Storage Account Type	Description
General Purpose v2 (GPv2)	Supports blobs, files, queues, tables. Best for most scenarios.
General Purpose v1 (GPv1)	Legacy option. Supports blobs, files, queues, tables.
Blob Storage	Optimized for storing large amounts of unstructured data (images, videos, documents, backups).
File Storage	Fully managed file shares in the cloud using SMB or NFS.
Queue Storage	Reliable messaging for decoupling applications and services.
Table Storage	Schema-less NoSQL store for structured data.
Disk Storage	Managed disks for Azure VMs (OS disk, Data disk).

☆ 2. Blob Storage – Key Concepts

- Stores data as objects (blobs) in containers.
- Three blob types : Block Blob, Append Blob, Page Blob.
- Ideal for backups, media, logs, static websites, data lakes.

Blob Hierarchy

Storage Account → Container → Blob

- Container : Logical grouping (like folders).
- Blob : The actual data (file).

Access Tiers (for Block Blob)

Frequent access, higher cost.	Infrequent access, lower cost.	Rare access, lowest cost.

Choose the tier based on how often your data is accessed.

☆ 3. File Storage (Azure Files)

- Fully managed file shares accessible via SMB (Windows) or NFS (Linux).
- Lift & shift applications with minimal code changes.
- Supports Azure AD authentication and access control.

Use Cases

- ✓ Application lift & shift
- ✓ Content sharing
- ✓ Home directories
- ✓ Log & data sharing
- ✓ Dev/Test environments

☆ 4. Queue Storage

- Stores messages in a queue that can be accessed by applications.
- Ensures reliable, asynchronous communication.
- Messages are kept until they are processed.

How Queue Storage Works

Application (Sender) → Send Message → Queue → Read Message → Application (Receiver)

☆ 5. Table Storage

- NoSQL store for structured, non-relational data.
- Best for large volumes of structured data with low latency.
- Data is stored in Tables → Entities → Properties.

Structure

Table → Entity → Properties

```
Name : Value
Age  : 25
City : Bangalore
...
```

☆ 6. Disk Storage

- Managed disks for Azure Virtual Machines.
- Types : Premium SSD, Standard SSD, Standard HDD, Ultra Disk.
- Supports snapshots for backup and restore.

Disk Types

Disk Type	Best For
Premium SSD	High IOPS workloads, databases, critical apps.
Standard SSD	Production workloads, web servers.
Standard HDD	Backup, infrequent access workloads.
Ultra Disk	Mission-critical, extremely high performance.

Storage Redundancy Options (Data Protection)

LRS (Locally Redundant Storage)	ZRS (Zone Redundant Storage)	GRS (Geo Redundant Storage)	RA-GRS (Read Access Geo Redundant Storage)	GZRS (Geo Zone Redundant Storage)
Data replicated 3 times within same datacenter.	Data replicated across multiple availability zones in a region.	Data replicated to a secondary region hundreds of miles away.	Same as GRS + read access to secondary region.	Data replicated across zones in primary region + geo replicated to secondary.

✓ Choose the right storage service for your data type and access pattern.

Right Storage,

6. Azure Networking Services

→ Azure networking services enable secure, reliable and high performance communication between resources, users and the internet.

☆ 1. Virtual Network (VNet)

- A VNet is a private network in Azure.
- It allows Azure resources to communicate with each other, the internet and on-premises networks securely.
- You can configure your own IP address ranges, subnets and routing.

VNet Components

Address Space | Subnets | Network Security Group (NSG) | Route Tables | Gateways (VPN/ExpressRoute)

☆ 2. Subnet

- Subnets divide the IP address space of a VNet.
- Each subnet can contain multiple Azure resources.
- You can associate NSG and route tables with a subnet.

Example

VNet Address Space : 10.0.0.0/16

Subnet-1	Subnet-2	Subnet-3
10.0.1.0/24	10.0.2.0/24	10.0.3.0/24

☆ 3. Network Security Group (NSG)

- NSG contains security rules that allow or deny inbound and outbound traffic.
- Rules are applied to subnets or network interfaces.

NSG Rule Example

Priority	Name	Direction	Source	Destination	Action	Port
100	Allow-HTTP	Inbound	Any	Any	Allow	80
200	Allow-SSH	Inbound	MyIP	Any	Allow	22
4096	Deny-All	Inbound	Any	Any	Deny	*

☆ 4. Public IP Address

- A public IP address is a unique IP on the internet.
- Used to access resources like VM, Load Balancer, Application Gateway etc.

Types

Static IP	Dynamic IP
Remains the same until released.	Changes when the resource is stopped.

☆ 5. Load Balancer

- Distributes incoming network traffic across multiple backend resources.
- Ensures high availability and scalability of applications.

Load Balancer Types

Internal Load Balancer	Public Load Balancer
• Used within a VNet	• Exposed to the internet
• For private traffic	• Has a public IP

☆ 6. Application Gateway

- Layer 7 load balancer (HTTP/HTTPS).
- Provides advanced routing, SSL termination, Web Application Firewall (WAF), URL based routing.

Key Features

- ✓ HTTP/HTTPS load balancing
- ✓ SSL offloading
- ✓ Web Application Firewall (WAF)
- ✓ Path based routing
- ✓ Health probes & session affinity

☆ 7. Azure DNS

- Host your domain in Azure.
- Provides name resolution using Microsoft's global anycast DNS network.

DNS Record Types

A	Maps domain to IPv4 address	CNAME	Alias of one domain to another
AAAA	Maps domain to IPv6 address	MX	Mail exchange records
TXT	Text records for verification		

☆ 8. Virtual Network Gateway

- Enables secure connectivity between VNet and on-premises networks.
- Types : VPN Gateway (Site-to-Site), ExpressRoute Gateway.

Connection Types

VPN (Site-to-Site)	ExpressRoute
Uses internet with IPSec encryption.	Uses private Microsoft network for dedicated connectivity.

Quick Recap

- ✓ Use VNet for private networking in Azure.
- ✓ Secure your resources with NSG and route tables.
- ✓ Use Application Gateway for Layer 7 traffic.
- ✓ Plan DNS and connectivity for global reach.

7. Azure Database Services

→ Azure provides a variety of fully managed database services to meet different application requirements. Key database offerings are:

☆ 1. Azure SQL Database

- Managed relational database service based on SQL Server.
- Supports most of the SQL Server features.
- Highly available, secure and scalable.

Deployment Options

Single Database	Elastic Pool	Managed Instance
Best for single database applications.	Share resources across multiple databases.	Full SQL Server compatibility in the cloud.

Use Cases

- ✓ Web applications
- ✓ Mobile backends
- ✓ eCommerce apps
- ✓ SaaS applications

☆ 2. Azure Cosmos DB

- Globally distributed, multi-model database service.
- Supports key-value, document, column-family and graph data.
- Built-in global distribution, high availability and low latency.

APIs Supported

SQL (Document)	MongoDB (Document)	cassandra (Column Family)	Gremlin (Graph)	Table (Key-Value)

Key Features

- ✓ Global distribution across any number of regions
- ✓ Five consistency models to balance performance & consistency
- ✓ Automatic indexing and scaling

☆ 3. Azure Database for MySQL

- Fully managed MySQL database in the cloud.
- Built on the latest stable community editions.
- High availability with built-in replication.

Deployment Options

Single Server	Flexible Server	Enterprise (Preview)
Best for dev/test & small apps.	Customizable, high availability, backups & scaling.	Mission critical apps with 99.99% availability.

Use Cases

- ✓ Web & mobile applications
- ✓ WordPress, Drupal, Joomla
- ✓ SaaS applications
- ✓ Data-driven applications

☆ 4. Azure Database for PostgreSQL

- Managed PostgreSQL database service.
- Supports open-source PostgreSQL community editions.
- Offers high performance, scalability and security.

Deployment Options

Single Server	Flexible Server
Ideal for dev/test & small workloads.	High availability, read replicas, private access, backups.

Key Features

- ✓ Automated backups and point-in-time restore
- ✓ Read replicas for high availability
- ✓ Private access with VNet integration

☆ 5. Azure Database for MariaDB

- Enterprise-grade MariaDB database as a service.
- Compatible with MySQL and optimized for high performance.
- Supports high availability and scaling.

Deployment Options

Single Server	Flexible Server
For dev/test & small apps.	High availability, read replicas, backups & scaling.

Use Cases

- ✓ Web applications
- ✓ SaaS applications
- ✓ eCommerce
- ✓ Analytics & reporting

☆ 6. Azure Cosmos DB Table API

- Fully managed NoSQL key-value store.
- Schema-less, high performance and low latency.
- Ideal for storing large amounts of structured data.

Characteristics

Schema-less NoSQL store	Low latency & high throughput	Automatic scaling	Global distribution

Use Cases

- ✓ Session storage
- ✓ Shopping carts
- ✓ User profiles
- ✓ IoT & telemetry data

Quick Recap

- ✓ Choose the right database based on your application needs.
- ✓ Consider data model, scalability, consistency and cost.

8. Azure Identity & Access Management (IAM)

→ Azure IAM helps you manage who can access your resources and what actions they can perform.

☆ 1. Microsoft Entra ID (Azure AD)

- Identity service to manage users, groups and applications.
- Provides authentication, authorization and directory services.
- Supports Single Sign-On (SSO) and Multi-Factor Authentication (MFA).

Key Features



User & Group Management



Single Sign-On (SSO)



Multi-Factor Authentication (MFA)

☆ 2. Users

- An identity representing a person or service.
- Can be assigned to groups and roles.

Types of Users



Member User (Access to resources)



Guest User (External users)

☆ 3. Groups

- Collection of users that can be managed as a single unit.
- Used to assign roles to multiple users easily.

Types of Groups



Microsoft 365 Groups



Security Groups



Distribution Groups

☆ 4. Roles in Azure

- Role defines a set of permissions.
- Roles can be assigned at different scopes.

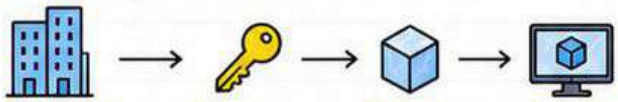
Built-in Role Examples

Role	Description
Owner	Full access to manage all resources.
Contributor	Can manage resources but cannot grant access.
Reader	Read-only access.
User Access Admin	Can manage user access to resources.
Billing Reader	Read-only access to billing information.

☆ 5. Scope of Role Assignment

- Roles can be assigned at different levels of scope.

Scope Levels (from wide to narrow)



Management Group → Subscription → Resource Group → Resource

☆ 6. Role Assignment

- Assign roles to users, groups or service principals.
- Can be done using Azure Portal, CLI, PowerShell or ARM templates.

Steps to Assign a Role

1. Choose the scope (Subscription / RG / Resource)
2. Select the role
3. Choose the user, group or service principal
4. Review + Assign



☆ 7. Service Principal

- Identity for applications, hosted services or automation tools.
- Used for non-interactive access to Azure resources.

Use Cases



Automation Scripts



Applications & APIs



CI/CD Pipelines

☆ 8. Privileged Identity Management (PIM)

- Helps manage, control and monitor access to critical roles.
- Provides just-in-time access and approval workflows.

PIM Capabilities

- ✓ Just-in-time role activation
- ✓ Approve access requests
- ✓ Access reviews & attestation
- ✓ Audit & monitoring





- ✓ Azure IAM is all about the right people having the right access.
- ✓ Use least privilege access — give minimum access required



9. Azure Monitor & Management Services

→ Azure Monitor helps you collect, analyze and act on telemetry data from your cloud and on-premises environments to ensure availability and performance of your applications.

☆ 1. Azure Monitor

- Collects data from Azure resources, applications and on-premises.
- Provides insights, dashboards, alerts and logs.
- Helps in troubleshooting and performance monitoring.

Capabilities

Metrics | Logs | Alerts | Dashboards | Insights

Use Cases

- ✓ Monitor application performance
- ✓ Detect and diagnose issues
- ✓ Set alerts and take action
- ✓ Visualize data with dashboards



☆ 2. Log Analytics Workspace

- Centralized place to store and analyze log data.
- Used by Azure Monitor, Sentinel, Automation and more.
- Based on Azure Monitor Logs.

How it Works

Data Sources → Collect → Store → Analyze → Visualize

☆ 3. Alerts in Azure Monitor

- Proactive notifications based on conditions.
- Can trigger actions like email, SMS, webhook, Logic App.

Alert Rule Components

Scope | Condition | Action Group | Severity

☆ 4. Action Groups

- Define a set of actions to be triggered by an alert.
- Reusable across multiple alerts.

Supported Actions

Email | SMS | Webhook | Logic App | Azure Function

☆ 5. Azure Advisor

- Personalized recommendations to improve reliability, security, performance and cost.
- Analyzes your Azure resources continuously.

Recommendation Categories

Security | Reliability | Performance | Cost | Operational Excellence

☆ 6. Azure Activity Log

- Provides insight into operations performed on resources.
- Tracks create, update, delete, read and other actions.

Captured Operations

Write | Delete | Read | Permissions | Others

☆ 7. Azure Service Health

- Provides real-time updates on the health of Azure services.
- Helps you understand and respond to issues.

Information Provided

Service Issues | Planned Maintenance | Health Advisories | Regional Status

☆ 8. Azure Automation

- Automates and orchestrates administrative tasks.
- PowerShell, Python, Runbooks and DSC support.

Key Features

Runbooks | Schedules | Hybrid Workers | Modules | Jobs & Logs

Azure Monitor Data Sources

Azure Resources | Virtual Machines | Applications | Network Devices | On-premises Servers | Databases | OS & Processes | Custom Sources

- ✓ Monitor everything, fix issues early.
- ✓ Review Advisor recommendations regularly.

10. Azure Security Services

→ Azure offers a wide range of security services to protect your cloud resources, data and applications.

☆ 1. Microsoft Defender for Cloud

- Unified security management and advanced threat protection across hybrid cloud workloads.
- Continuous assessment of security posture.
- Workload protection, vulnerability management and recommendations.

Key Features

Security Posture Management | Threat Protection | Regulatory Compliance | Workload Protection | Vulnerability Management

Use Cases

- ✓ Secure VMs, containers, databases and apps
- ✓ Detect misconfigurations and threats
- ✓ Meet compliance standards



☆ 2. Azure Key Vault

- Safely store and manage secrets like keys, passwords, certificates, and more.
- Support for key rotation and access policies.

Stores

Keys | Secrets | Certificates | Access Policies | Key Rotation

Use Cases

- ✓ Store DB passwords, API keys, certificates
- ✓ Secure access with RBAC and access policies
- ✓ Integrate with apps and services



☆ 3. Azure Firewall

- Cloud-native network security service.
- Protects your Azure resources with stateful firewall as a service.
- Supports inbound and outbound traffic filtering.

Key Capabilities

Traffic Filtering | Threat Intelligence | Outbound Filtering | Network Segmentation | Logging & Monitoring

Use Cases

- ✓ Secure hub-spoke network architecture
- ✓ Control inbound and outbound traffic
- ✓ Centralized threat protection and logging



☆ 4. Azure DDoS Protection

- Protects applications from DDoS attacks.
- Always-on network layer protection.
- Automatic mitigation with minimal configuration.

Protection Plans

Basic (Free)
- Always-on Network Protection

Standard
- Advanced mitigation
- Telemetry & alerts
- Cost protection

Use Cases

- ✓ Protect public-facing applications
- ✓ Ensure high availability
- ✓ Mitigate volumetric attacks



☆ 5. Azure Sentinel (SIEM & SOAR)

- Cloud-native SIEM (Security Information and Event Management) solution.
- Collects data, detects threats and automates response.

Key Features

Threat Detection | Security Analytics | Automated Response | Integration (100+ Connectors) | Incident Management

Use Cases

- ✓ Detect advanced threats
- ✓ Investigate security incidents
- ✓ Automate threat response



☆ 6. Azure Policy

- Enforce organizational standards and assess compliance at scale.
- Define rules and policies for your resources.

Policy Components

Definitions (What to enforce) | Assignments (Where to apply) | Initiatives (Group of policies) | Compliance Monitoring | Remediation (Auto fix non-compliance)

Use Cases

- ✓ Enforce tagging, location, naming standards
- ✓ Ensure compliance with regulations
- ✓ Prevent non-compliant resource creation



Quick

- ✓ Use Defender for Cloud to secure workloads.
- ✓ Store secrets safely in Key Vault.
- ✓ Detect threats early with Sentinel.
- ✓ Enforce rules with Azure Policy.



11. Azure Storage Services

→ Azure Storage provides highly durable, scalable, and secure solutions for storing data in the cloud.

☆ 1. Azure Blob Storage

- Object storage solution for unstructured data.
- Stores text or binary data like images, videos, documents, backups, etc.
- Highly scalable and cost-effective.

Key Features

Massive Scalability | High Durability | High Availability | Access Tier (Hot, Cool, Archive) | Secure Access

Use Cases

- ✓ Store images, videos, audio files
- ✓ Backup and disaster recovery
- ✓ Data analytics and big data



☆ 2. Azure File Storage

- Fully managed file shares in the cloud.
- Accessible via SMB (Server Message Block) or NFS protocols.
- Lift-and-shift legacy applications to the cloud.

Key Features

SMB / NFS Access | Cloud Backup | Enterprise Security | Scalable Capacity | High Performance

Use Cases

- ✓ Legacy applications
- ✓ Shared content and media
- ✓ Home directories for cloud users



☆ 3. Azure Queue Storage

- Stores large numbers of messages that can be accessed from anywhere in the world.
- Decouples and scales applications.

Key Features

Message Storage | Reliable Messaging | TTL Support | Scalable & Durable | Access from Anywhere

Use Cases

- ✓ Background task processing
- ✓ Decouple application components
- ✓ Store messages for later processing



☆ 4. Azure Table Storage

- NoSQL key-attribute store.
- Stores structured, schemaless data.
- Designed for massive scale and low cost.

Key Features

Schema-less Design | High Scalability | Cost Effective | Global Distribution | Secure Access

Use Cases

- ✓ Storing structured NoSQL data
- ✓ Catalogs, user profiles, session data
- ✓ IoT and analytics data



☆ 5. Azure Disk Storage

- Provides virtual hard disks (VHDs) for Azure VMs.
- Managed Disks for high performance and scalability.
- Supports SSD and HDD disk types.

Key Features

Managed Disks | High Performance | Data Redundancy | Snapshots | Encryption

Use Cases

- ✓ OS and data disks for VMs
- ✓ High performance applications
- ✓ Backups and disaster recovery



☆ 6. Azure Data Lake Storage Gen2 (ADLS Gen2)

- Built for big data analytics.
- Combines the scalability of Blob Storage with the performance of Data Lake.
- Hierarchical namespace and POSIX-like access control.

Key Features

High Scalability | Hierarchical Namespace | Fine-grained Security | Optimized for Analytics | Integration with Services

Use Cases

- ✓ Big data analytics
- ✓ Machine learning workloads
- ✓ Data warehousing and data engineering



Quick

- ✓ Choose the right storage based on your data type and access pattern.
- ✓ Use access tiers to optimize storage cost.
- ✓ Integrate storage with Azure services for

12. Azure Networking Services

→ Azure Networking services help you build secure, scalable and reliable network infrastructure in the cloud.

☆ 1. Virtual Network (VNet)

- Foundation for your private network in Azure.
- Enables secure communication between resources.
- You can create subnets, route tables, NSGs and connect to on-premises.

Key Features



Use Cases

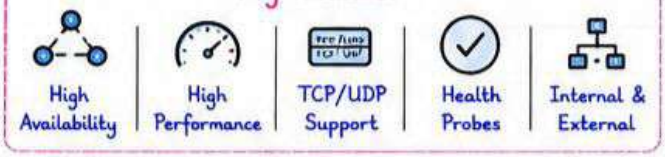
- ✓ Host Azure resources securely
- ✓ Connect to on-premises networks
- ✓ Micro-segmentation of workloads



☆ 2. Azure Load Balancer

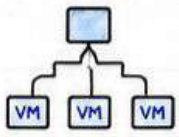
- Distributes incoming traffic across multiple VMs for high availability.
- Supports TCP, UDP and inbound traffic.
- Operates at Layer 4 (Transport layer).

Key Features



Use Cases

- ✓ Distribute network traffic
- ✓ Improve application availability
- ✓ Handle large-scale traffic



☆ 3. Azure Application Gateway

- Web traffic load balancer (Layer 7).
- Provides advanced routing, SSL termination, WAF and more.
- Ideal for web applications.

Key Features



Use Cases

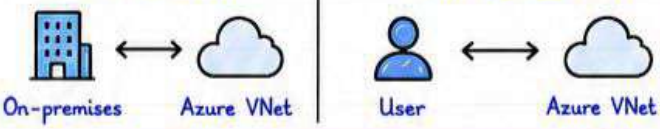
- ✓ Host and secure web applications
- ✓ Route traffic based on URL
- ✓ Protect apps from web attacks



☆ 4. Azure VPN Gateway

- Enables secure connection between on-premises networks and Azure VNets over the internet.
- Supports Site-to-Site and Point-to-Site VPN.

VPN Types



Use Cases

- ✓ Secure hybrid connectivity
- ✓ Remote access for users
- ✓ Connect branch offices



☆ 5. Azure DNS

- Microsoft's DNS service to host and manage DNS domains.
- Provides name resolution using high availability and global anycast network.

Key Features



Use Cases

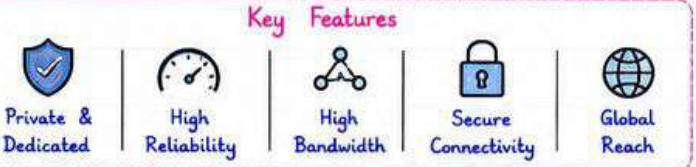
- ✓ Host public DNS zones
- ✓ Manage domain records
- ✓ Improve application availability



☆ 6. Azure ExpressRoute

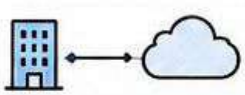
- Provides private, dedicated and reliable connection between Azure and on-premises.
- Not over the public internet.

Key Features



Use Cases

- ✓ Connect data centers to Azure
- ✓ Avoid internet dependency
- ✓ Support hybrid cloud solutions



13. Azure Compute Services

→ Azure Compute services provide virtual machines, containers, serverless and batch processing capabilities to run your applications in the cloud.

☆ 1. Azure Virtual Machines (VM)

- Provision secure, resizable Windows or Linux VMs in minutes.
- Full control over OS, network and storage.
- Ideal for lift-and-shift and custom applications.

Key Features

Scalable Compute | Security & Compliance | Managed Disks | High Availability | Monitoring & Alerts

Use Cases

- ✓ Web and application servers
- ✓ Development and test environments
- ✓ Database and enterprise applications



☆ 2. Azure App Service

- Fully managed platform to build, deploy and scale web apps, APIs and mobile backends.
- Supports .NET, Java, Node.js, Python, PHP and more.

Key Features

Web Apps | Easy Deployment | Built-in Security | CI/CD Integration | Auto Scaling

Use Cases

- ✓ Hosting websites and APIs
- ✓ RESTful APIs and backends
- ✓ Mobile app backends



☆ 3. Azure Container Instances (ACI)

- Run containers without managing servers.
- Ideal for small, short-lived container workloads.
- Pay only for the resources you use.

Key Features

Serverless Containers | Quick Deployment | Cost Effective | VNet Integration | Secure by Default

Use Cases

- ✓ Microservices and containerized apps
- ✓ Batch jobs and task automation
- ✓ Event-driven workloads



☆ 4. Azure Functions

- Serverless compute that runs code in response to events.
- No infrastructure management.
- Pay only for execution time.

Key Features

Event Driven | Multiple Languages | Auto Scaling | Pay per Execution | Easy Integration

Use Cases

- ✓ Real-time file processing
- ✓ IoT data processing
- ✓ Automation and scheduled tasks



☆ 5. Azure Batch

- Run large-scale parallel and high performance computing workloads.
- Automatically scales compute resources.
- Ideal for HPC and big data processing.

Key Features

Parallel Processing | Dynamic Scaling | Job Scheduling | Cost Efficient | Monitoring & Logs

Use Cases

- ✓ Scientific simulations
- ✓ Rendering and encoding
- ✓ Financial modeling



☆ 6. Azure Kubernetes Service (AKS)

- Managed Kubernetes service to deploy, manage and scale containerized applications.
- Simplifies container orchestration.
- Supports DevOps and CI/CD.

Key Features

Managed Kubernetes | Security & Compliance | Auto Scaling | DevOps Friendly | Monitoring Built-in

Use Cases

- ✓ Containerized applications
- ✓ Microservices architectures
- ✓ DevOps and CI/CD pipelines



14. Azure Database Services

→ Azure Database services help you store, manage and scale data with security, high availability and performance for all types of applications.

☆ 1. Azure SQL Database

- Fully managed relational database in the cloud.
- Built-in intelligence, high availability and security.
- Automatic backups, patching and scaling.

Key Features

Fully Managed | High Availability | Intelligent Performance | Auto Scaling | Security Built-in

Use Cases

- ✓ Web applications and SaaS
- ✓ Business applications (ERP, CRM)
- ✓ Reporting and analytics



☆ 2. Azure Database for MySQL

- Managed MySQL database service.
- Supports high availability, backups and scaling.
- Compatible with MySQL community edition.

Key Features

MySQL Compatible | High Availability | Auto Backup | Scale Up/Down | Secure Connection

Use Cases

- ✓ Web and mobile applications
- ✓ Content management systems
- ✓ E-commerce platforms



☆ 3. Azure Cosmos DB

- Globally distributed, multi-model database service.
- Supports key-value, document, column, graph.
- Built for massive scale and low latency.

Key Features

Global Distribution | Multi-model Database | Low Latency | Automatic Partitioning | High Throughput

Use Cases

- ✓ IoT applications
- ✓ Gaming and personalization
- ✓ Real-time analytics



☆ 4. Azure Database for PostgreSQL

- Managed PostgreSQL database in the cloud.
- Enterprise-grade security and performance.
- Supports extensions and open-source compatibility.

Key Features

PostgreSQL Compatible | High Availability | Auto Backup | Scalable Compute | Enterprise Security

Use Cases

- ✓ Enterprise applications
- ✓ GIS and spatial applications
- ✓ Analytics and reporting



☆ 5. Azure Synapse Analytics

- Unlimited analytics service that brings together data integration, enterprise data warehousing and big data analytics.
- Analyze data at scale and get insights faster.

Key Features

Unified Analytics | Data Warehousing | Big Data Analytics | Data Integration | Real-time Insights

Use Cases

- ✓ Data warehousing and BI
- ✓ Big data analytics
- ✓ Data lake analytics



☆ 6. Azure Redis Cache

- In-memory data store for high performance.
- Improves application performance and scalability.
- Supports pub/sub messaging.

Key Features

High Performance | Scalability On-demand | Pub/Sub Messaging | Data Persistence | Secure Access

Use Cases

- ✓ Session store
- ✓ Caching
- ✓ Real-time leaderboards



15. Azure DevOps Services

→ Azure DevOps helps teams plan, code, build, test and deploy applications using a set of collaborative tools and services.

★ 1. Azure Boards

- Plan, track and discuss work across your team.
- Agile tools like Kanban boards, backlogs, dashboards and reports.
- Improve visibility and collaboration.

Key Features

Work Tracking | Backlogs | Dashboards & Reports | Team Collaboration | Agile Planning

Use Cases

- ✓ Agile project management
- ✓ Track bugs, tasks and user stories
- ✓ Monitor team progress and velocity

★ 2. Azure Repos

- Git repositories in the cloud.
- Unlimited private repos and secure collaboration.
- Code review, branching and policies.

Key Features

Git Repos | Branching & Merging | Code Review | Security & Policies | Access Control

Use Cases

- ✓ Store and manage source code
- ✓ Collaborate using pull requests
- ✓ Version control and code history

★ 3. Azure Artifacts

- Create, host and share packages with your team.
- Supports NuGet, Maven, npm, Python and more.
- Integrated with Azure Pipelines.

Key Features

Package Management | Multiple Formats | Feed Sharing | Upstream Sources | Pipeline Integration

Use Cases

- ✓ Host private packages
- ✓ Share libraries across projects
- ✓ Manage dependencies securely

★ 4. Azure Pipelines

- Continuous Integration and Continuous Delivery (CI/CD).
- Build, test and deploy code to any cloud or on-premises.
- Supports multiple languages and platforms.

Key Features

CI/CD Automation | Multi-language Support | Cloud & On-prem Deployment | Security & Compliance | Built-in Analytics

Use Cases

- ✓ Automate build, test and release pipelines
- ✓ Deploy to Azure, other clouds or servers
- ✓ Continuous delivery of applications

★ 5. Azure Test Plans

- Plan and track all your testing activities.
- Create test cases, run tests and log bugs.
- Improve quality with traceability and reports.

Key Features

Test Case Management | Bug Tracking | Test Runs | Requirement Traceability | Test Reports

Use Cases

- ✓ Plan and organize testing activities
- ✓ Track bugs and test results
- ✓ Ensure quality and release confidence

★ 6. Azure DevOps Extensions

- Extend and customize Azure DevOps to fit your needs.
- Hundreds of extensions available in the marketplace.
- Improve productivity and add new capabilities.

Key Features

Marketplace Extensions | Custom Features | Boost Productivity | Secure & Reliable | Easy Integration

Use Cases

- ✓ Add new features and tools
- ✓ Integrate with third-party services
- ✓ Customize workflows and dashboards